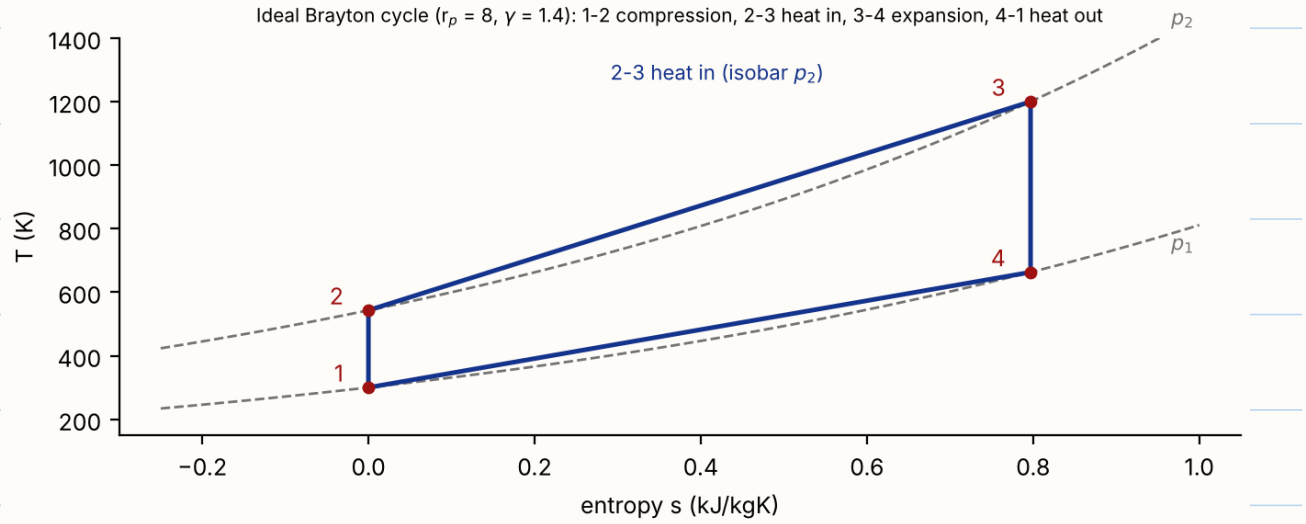
UNIT 2: Gas Turbine Cycle

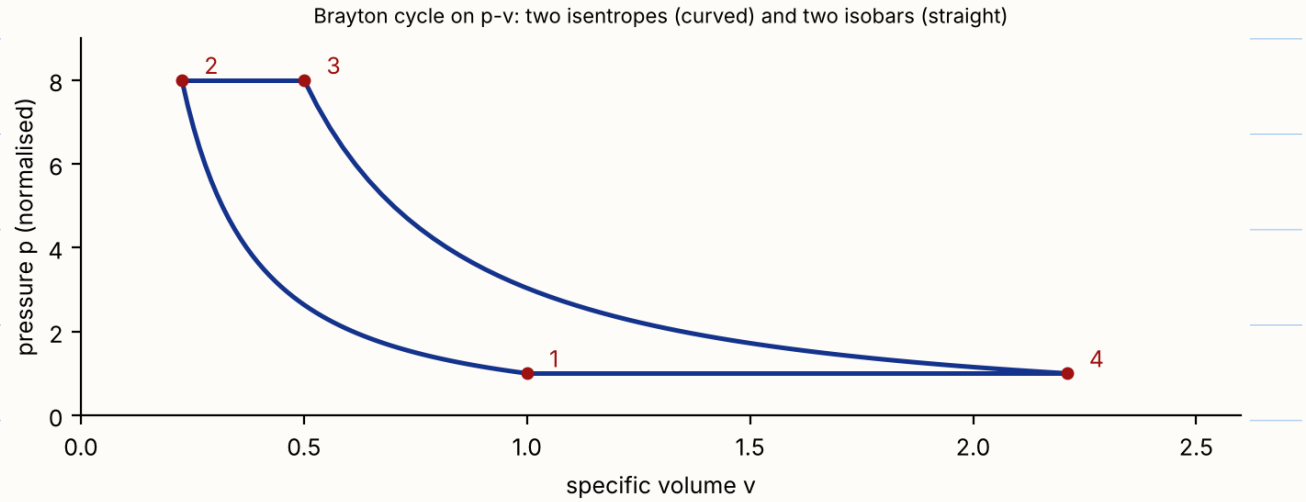
Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Brayton cycle; regenerative cycle; reheat; inter-cooling; turboprop, turbojet and turbofan engines; thrust augmentation; examples.

1. The ideal Brayton cycle

Brayton cycle on T-s and p-v





The Brayton cycle is the ideal cycle for gas turbines. It has four processes:

- 1–2: isentropic compression in the compressor.
- 2–3: constant-pressure heat addition in the combustor.
- 3–4: isentropic expansion in the turbine.
- 4–1: constant-pressure heat rejection (exhaust to atmosphere).

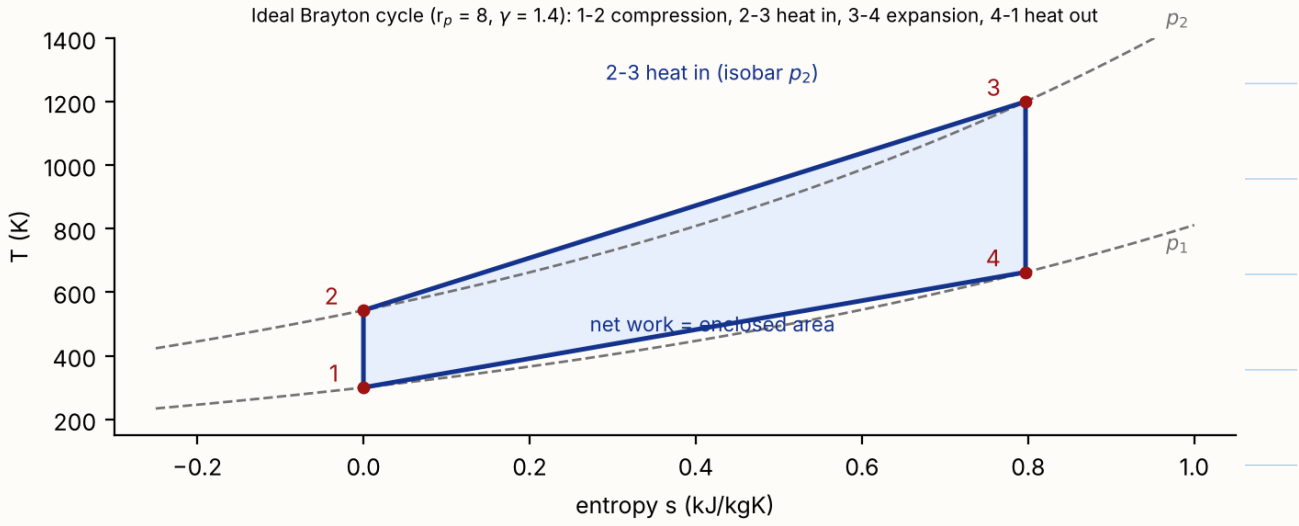
On a T-s diagram the cycle is two isobars ($p_2 = p_3$, $p_1 = p_4$) joined by two isentropes. On a p-v diagram it is a closed loop. Net work equals the area enclosed.

Pressure ratio $r_p = p_2 / p_1$ ⇒ $\eta_{th} = 1 - 1 / r_p^{(\gamma-1)/\gamma}$

Important result: for the ideal cycle, efficiency depends only on the pressure ratio. It does not depend on the turbine inlet temperature.

2. Work and back-work ratio

Work as areas on the T-s diagram



$w_c = c_p(T_2 - T_1) = c_p T_1(r_p^{(\gamma-1)/\gamma} - 1)$ $w_t = c_p(T_3 - T_4)$

$w_{net} = w_t - w_c$

The **back-work ratio** is w_c / w_t. Gas turbines have a large back-work ratio (often 40–80%) because compression consumes most of the turbine work. This is why improving compressor and turbine efficiency matters so much.

Maximum net work occurs at an optimum pressure ratio. For a given maximum and minimum temperature (T_3 and T_1):

$$r_{p,opt} = (T_3 / T_1)^{\gamma / 2(\gamma-1)}$$

Example: *For $r_p = 8$ and $\gamma = 1.4$, find the ideal Brayton efficiency.*

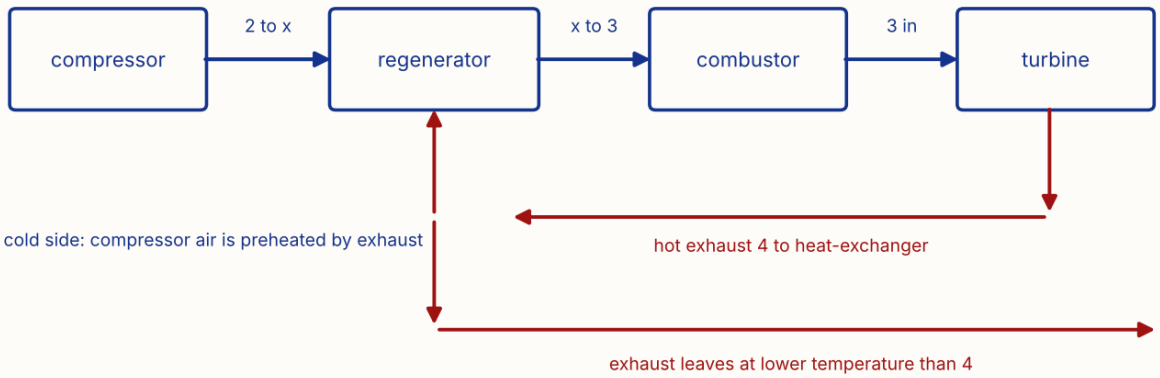
Solution: $r_p^{(\gamma-1)/\gamma} = 8^{0.2857} = 1.811$, so $\eta_{th} = 1 - 1/1.811 = 0.448$, i.e. about 44.8%.

3. Real cycle losses

Real components are not isentropic. The compressor efficiency and turbine efficiency reduce the net work. Combustor pressure loss (typically 3–6% of inlet pressure) and the mechanical losses also matter. Turbine inlet temperature (TIT) is limited by materials and cooling, so TIT is the main design variable for improving output.

4. Regenerative cycle

Regenerative cycle layout



A regenerator (heat exchanger) uses hot exhaust gas to preheat the compressor delivery air before combustion. Heat added in the combustor is reduced, so the cycle efficiency rises.

Effectiveness $\varepsilon = (T_x - T_2) / (T_4 - T_2)$ $\quad$ **$q_{in} = c_p(T_3 - T_x)$**

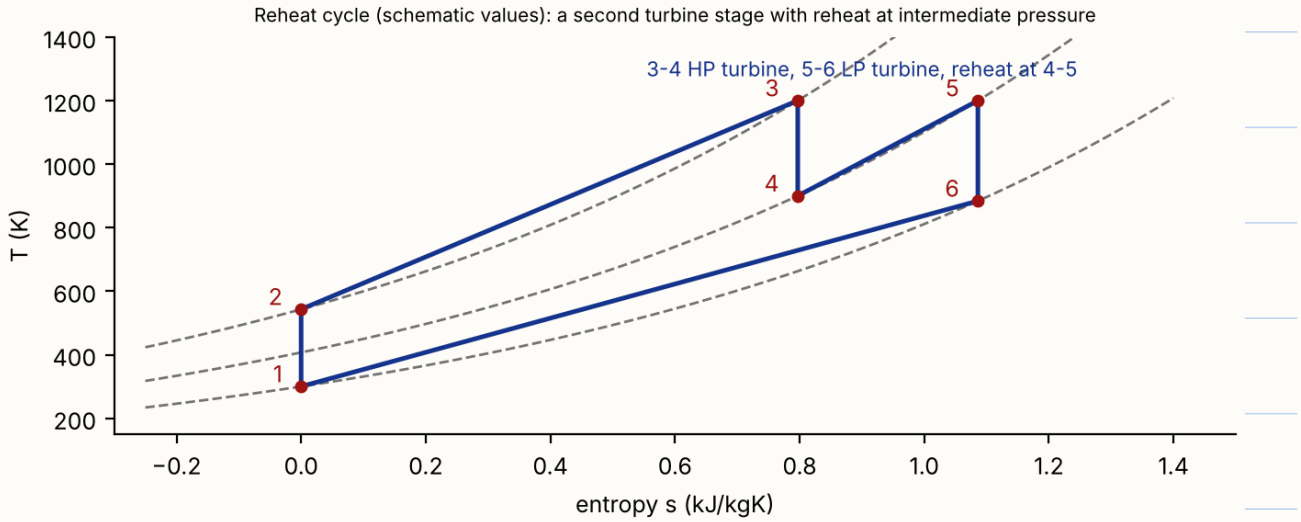
Here T_x is the air temperature leaving the regenerator on the compressor side. An ideal regenerator has $\varepsilon = 1$ and $T_x = T_4$.

- Regeneration is most useful at **low pressure ratios**, where the exhaust is still hot compared with the compressor delivery.
- At high pressure ratios the compressor outlet is already hotter than the exhaust, so the regenerator gives no benefit.

- The regenerator adds pressure loss and cost, so it is a trade-off with the efficiency gain.

5. Reheat and inter-cooling

Reheat cycle on T-s (schematic values)



Inter-cooling: compression is split into two stages with a cooler between them. Cooling reduces the specific volume of air, so less work is needed for the second stage. For two stages with the same pressure ratio per stage, the minimum compressor work occurs when:

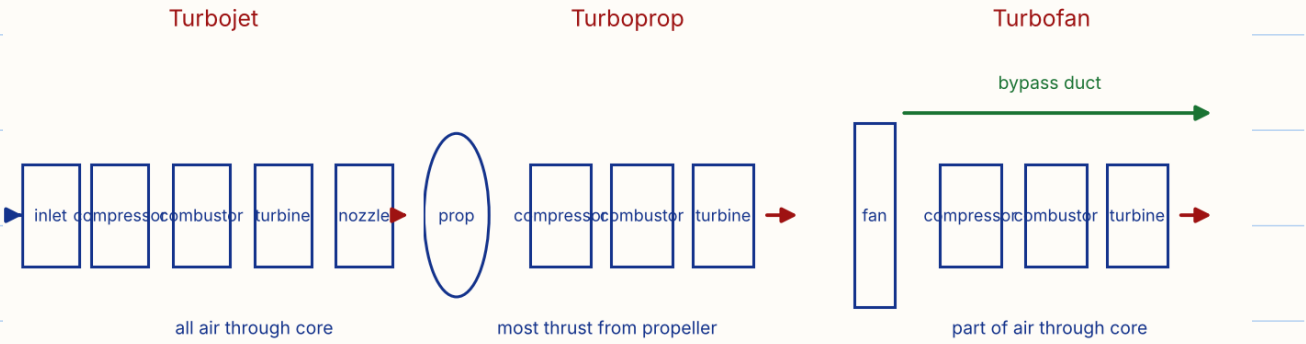
Intermediate pressure $p_i = \sqrt{(p_1\ p_2)}$ (equal pressure ratio per stage)

Reheat: expansion is split into two turbines with a second combustor between them. The gas is reheated before the second turbine. Turbine

work increases, so net work increases for the same compressor work. Intercooling and reheat together increase the overall net work but add complexity. Regeneration, intercooling and reheat are often combined to get high efficiency.

6. Aero engines: turbojet

Turbojet, turboprop and turbofan



In a turbojet, all air passes through the core. The gas leaves the nozzle at high velocity and the reaction gives thrust. The main thrust equation is:

$$F = \dot{m}\,(V_e - V_0) + (p_e - p_0)\,A_e$$

Here V_0 is flight speed, V_e is the jet exit velocity, p_e and p_0 are the exit and ambient pressures, and A_e is the nozzle exit area. If the nozzle is

fully expanded ($p_e = p_0$), only the momentum term remains.

Propulsive efficiency is the ratio of useful propulsive power to the rate of kinetic energy given to the jet:

$$\eta_p = 2\,V_0 / (V_e + V_0) \qquad \eta_0 = \eta_{th} \times \eta_p$$

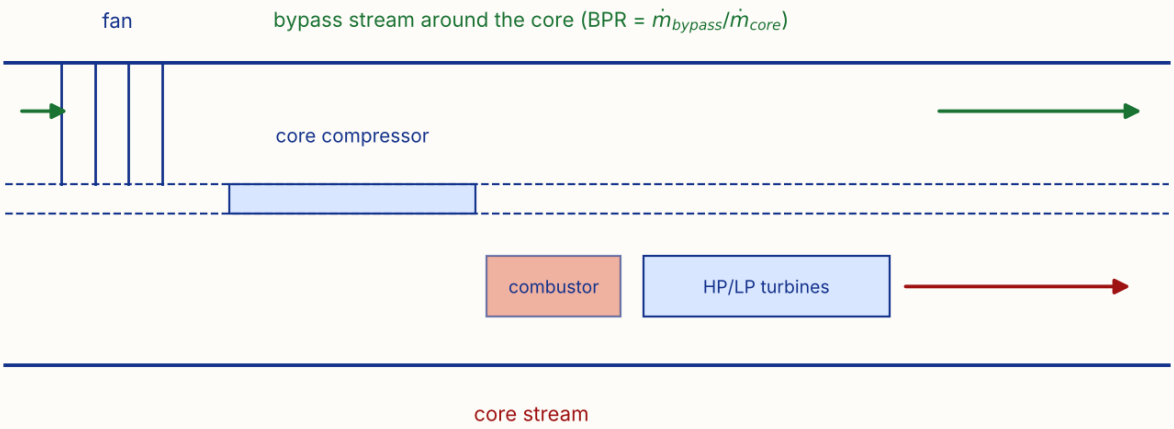
Propulsive efficiency is high when V_e is close to V_0. The turbojet has a high jet velocity, so it is efficient only at high speed.

7. Turboprop

In a turboprop, the gas generator drives a propeller through a reduction gearbox. Most of the thrust comes from the propeller, which moves a large mass of air at low velocity. This gives high propulsive efficiency at low to medium speeds (up to about 0.8 Mach), and it is ideal for regional transport.

8. Turbofan

Turbofan cross-section



A fan at the front pushes air through a bypass duct around the core.

The bypass ratio is:

$$\mathbf{BPR = \dot{m}_{bypass} / \dot{m}_{core}}$$

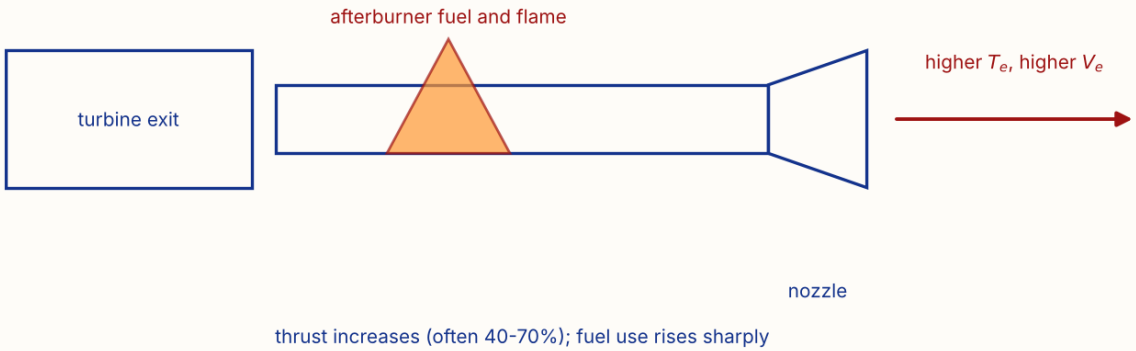
- Large BPR (high-bypass turbofan): low jet velocity, high propulsive efficiency, lower fuel consumption and lower noise.
- Low BPR (low-bypass turbofan): higher jet velocity, used for supersonic and military applications.
- Turbofans cover most civil aviation today because they combine the efficiency of the propeller with the speed of the turbojet.

Engine	Air through core?	Main thrust source	Best speed range

Turbojet	All air	High-velocity jet	High speed
Turboprop	All air	Propeller	Low to medium speed
Turbofan	Part of air	Fan + jet	Medium to high speed

9. Thrust augmentation (afterburner / reheat)

Afterburner in the jet pipe



thrust increases (often 40–70%); fuel use rises sharply

Fuel is injected into the hot jet pipe downstream of the turbine and burned. This raises the exit temperature and velocity, so the thrust increases (often by 40–70%). The cost is a large increase in specific fuel consumption, so afterburners are used mainly for take-off and combat. Water or methanol injection into the compressor is another way to boost thrust for a short time.

10. Worked examples

Example: *A turbojet has $\dot{m}$ = 50 kg/s, flight speed V_0 = 250 m/s, exit velocity V_e = 600 m/s and $p_e = p_0$. Find the thrust and propulsive efficiency.*

Solution: F = 50 × (600 − 250) = 17.5 kN. η_p = 2 × 250 / (600 + 250) = 500 / 850 = 0.588.

Example: *A regenerator has ε = 0.8, T_2 = 500 K and T_4 = 800 K. Find T_x.*

Solution: $T_x = T_2 + \varepsilon (T_4 - T_2)$ = 500 + 0.8 × 300 = 740 K.

11. Exam quick points

- Ideal Brayton efficiency depends only on pressure ratio: $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$.
- Back-work ratio of a gas turbine is high, so compressor efficiency is critical.
- Regenerator helps at low pressure ratio. Intercooling uses equal pressure ratios for minimum work.
- Thrust = $\dot{m}(V_e - V_0) + (p_e - p_0)A_e$. High BPR turbofan gives high η_p.

12. Practice questions

Q1. Draw the T-s diagram of the ideal Brayton cycle and label all four processes.

Q2. Explain why regeneration is less useful at high pressure ratio.

Q3. A turbofan has BPR = 5, core flow 40 kg/s and V_0 = 230 m/s. Explain qualitatively how the thrust is shared between the fan and core.